

DATA PRIVACY AND SECURITY

Prof. Daniele Venturi

Chapter 0: Course Overview

**Master's Degree in Data Science
Sapienza University of Rome**



About Myself

- Full Professor at the **Computer Science** Department
- Research focus: Theoretical and applied **cryptography**
- Personal homepage (contact info, research topics, office hours, etc.):

<https://dventuri83.github.io>

- Web page for this course:

https://dventuri83.github.io/teaching/2_dps/



Logistic

- The lectures are offered **exclusively** in person
 - No recordings will be available
 - Active participation is **highly** recommended
- Course material: Slides and bibliographic references from the course homepage



Exams

- **Oral exam** on the topics covered in class
- Students' **projects**
 - Choose a topic **during the semester**
 - Collaboration between students is **encouraged**
- Final grade: Oral exam (70%) and project presentation (30%)
- Exams sessions (plenary): January, February, June, July, and September



Syllabus

- Introduction to **cryptography**
 - **Symmetric** and **asymmetric** cryptography, **key exchange** protocols, **post-quantum** cryptography
- Differential Privacy
 - **Privacy-preserving statistics** on datasets
- Proof systems
 - Interactive proofs, **zero knowledge**, SNARKs
- Cryptocurrencies and **distributed ledgers**
 - Bitcoin, altcoins
- Secure **multiparty computation**
 - Garbled circuits, PIR, PSI



Bibliography

- J. Katz, Y. Lindell. *“Introduction to Modern Cryptography.”* Chapman & Hall, 3rd Edition
- Y. Lindell (Editor). *“Tutorials on the Foundations of Cryptography.”* Springer
- A. Chiesa, E. Yogev. *“Building Cryptographic Proofs from Hash Functions”* Springer
- E. Shi. *“Foundations of Distributed Consensus and Blockchains”*. Preliminary draft
- C. Hazay, Y. Lindell. *“Efficient Secure Two-party Protocols”*. Springer

